

QC02_all_wrong

Every compared global-embedding system answers incorrectly. Development set, train_40k, seed 0. Condition: normal.

question-only head: a classifier over the frozen CLIP question embedding alone, with no image input; concatenation head: a classifier over the frozen CLIP image and question embeddings concatenated; fusion head: the concatenation plus the elementwise product and absolute difference of the two embeddings.



image 2385131

Question: What is the person to the left of the taxi cab wearing?

Gold answer: jeans

Question-only: shirt

✗ incorrect

Concatenation: truck

✗ incorrect

Fusion: truck

✗ incorrect

Illustrative only. A shared failure. It does not identify a cause and must not be captioned as a reasoning failure without independent evidence.



image 2384279

Question: What color do you think the water has?

Gold answer: gray

Question-only: blue

✗ incorrect

Concatenation: brown

✗ incorrect

Fusion: brown

✗ incorrect

Illustrative only. A shared failure. It does not identify a cause and must not be captioned as a reasoning failure without independent evidence.

QC03_fusion_right_concat_wrong

Fusion correct where concatenation is wrong. Development set, train_40k, seed 0. Condition: normal.

concatenation head: a classifier over the frozen CLIP image and question embeddings concatenated; fusion head: the concatenation plus the elementwise product and absolute difference of the two embeddings.



image 2320225

Question: On which side is the red vehicle?

Gold answer: right

Concatenation: left

✗ incorrect

Fusion: right

✓ correct

Illustrative only. One row where the two systems disagree. It illustrates the aggregate contrast; it is not evidence for it, and the aggregate contrast is capacity-confounded.



image 2364920

Question: What color does the animal that the lamb is below have?

Gold answer: brown

Concatenation: white

✗ incorrect

Fusion: brown

✓ correct

Illustrative only. One row where the two systems disagree. It illustrates the aggregate contrast; it is not evidence for it, and the aggregate contrast is capacity-confounded.

QC05_question_side_pretraining

The pretrained question encoder A1 correct where its random control A1r is wrong. Development set, train_40k, seed 0. Condition: normal.

A1: the pretrained frozen SmoLM2-135M question encoder feeding the latent-query reasoner; A1r: the architecture-matched randomly initialised SmoLM2-135M question encoder, the within-size control for A1.



image 2393299

Question: Do you see either cookbooks or rugs?

Gold answer: no

A1 (pretrained): no

✓ correct

A1r (random): yes

✗ incorrect

Illustrative only. Illustrates the only clean within-size question-side pretraining contrast; a single row is not evidence of the effect.



image 2387771

Question: Is the drawer to the right of the stove?

Gold answer: yes

A1 (pretrained): yes

✓ correct

A1r (random): no

✗ incorrect

Illustrative only. Illustrates the only clean within-size question-side pretraining contrast; a single row is not evidence of the effect.

QC07_visual_reliance

Fusion correct under the normal image and wrong under the shuffled image. Development set, train_40k, seed 0. Condition: normal and shuffled, paired on the same question.

fusion head reading the question's own image; the same fusion head, with the image embedding replaced by another development image's under the recorded permutation.



image 2402937
the question's own image



image 2407981
the image that replaced it

Question: What's the plant on?

Gold answer: shelf

Fusion, real image: shelf

✓ correct

Fusion, wrong image: bed

✗ incorrect

Illustrative only. Demonstrates reliance on the image for this row. It is not proof of grounding, and the wrong-image partner must be shown so a reader can see what replaced it.



image 2410905
the question's own image



image 2385812
the image that replaced it

Question: Is the truck to the right or to the left of the motorcycle that is on the street?

Gold answer: right

Fusion, real image: right

✓ correct

Fusion, wrong image: left

✗ incorrect

Illustrative only. Demonstrates reliance on the image for this row. It is not proof of grounding, and the wrong-image partner must be shown so a reader can see what replaced it.

QC09_deep_reasoning_failure

A question with four or more program steps that fusion answers incorrectly. Development set, train_40k, seed 0. Condition: normal.
fusion head: the concatenation plus the elementwise product and absolute difference of the two embeddings.



image 2358266

Question: Are there any helmets or hearts in this scene?
Gold answer: yes
GQA program steps: 5
Fusion: no

✗ incorrect

Illustrative only. Illustrates where the pooled four-or-more-step deficit lives. It does not show the model failed BECAUSE the question was deep; no causal attribution may be made from an example.

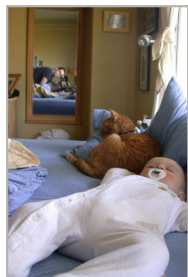


image 2361055

Question: Who is reflected in the mirror on the left side of the picture?
Gold answer: man
GQA program steps: 4
Fusion: left

✗ incorrect

Illustrative only. Illustrates where the pooled four-or-more-step deficit lives. It does not show the model failed BECAUSE the question was deep; no causal attribution may be made from an example.